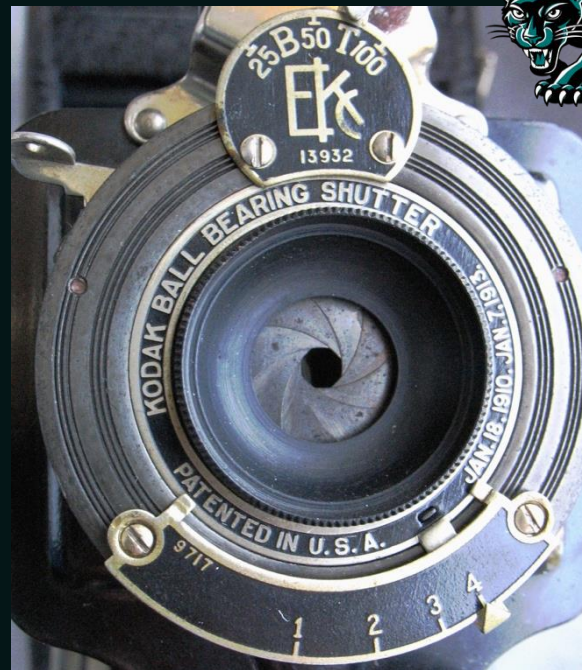


APERTURE

PART 1: HOW APERTURE WORKS

The opening inside your lens, the f-stop numbers, and what they do to your photo.

CANON EOS R50 & 50MM RF 1.8





WHAT YOU WILL LEARN

OUTCOMES / 01 / 20

01

NAME THE PARTS THAT CONTROL LIGHT

You will know what the aperture is and where it sits inside your lens.

02

READ ANY F-STOP NUMBER

You will look at f/1.8 or f/11 and know right away how open the lens is.

03

USE THE BACKWARDS RULE

You will remember that a small number means a big opening, every time.

04

SET APERTURE ON THE R50

You will use manual mode and set aperture, shutter, and ISO.

05

KNOW WHAT YOUR LENSES CAN DO

You will know the widest aperture on each lens in our kit.



Aperture is one of three controls that decide how a photo looks. Part 1 is about light. Part 2 is about focus.



Aperture

The opening inside your lens that lets light through. You can make it wider or smaller.

F-stop

The number that tells you how wide the aperture is open. It is written like f/4 or f/5.6.

Diaphragm

The ring of thin metal blades inside the lens. They slide open and closed to change the aperture.

Stop

One full step of light. Move one stop and you either double the light or cut it in half.

These four words show up in every camera manual, every lens review, and every job on a set. Learn them now and the rest of the unit reads easily.



Maximum aperture

The widest a lens can open. It is printed right on the front of the lens.

Fast lens

A lens with a very wide maximum aperture, like f/1.8. It lets in a lot of light.

Variable aperture

A zoom lens whose widest opening gets smaller as you zoom in. Most kit lenses work this way.

Depth of field

How much of the photo looks sharp from front to back. Part 2 covers this in full.

Maximum aperture is the number that decides what a lens can do in a dark room. It is the first thing a photographer checks before picking a lens for a job.



WHAT IS APERTURE?

CONTENT / 04 / 20

Think about the pupil in your eye. In a dark room it opens wide. In bright sun it shrinks. Your lens does the same thing.

01

A HOLE YOU CONTROL

The aperture is an opening inside the lens. You decide how wide it opens.

02

IT SETS HOW MUCH LIGHT GETS IN

A wider opening lets more light reach the sensor.

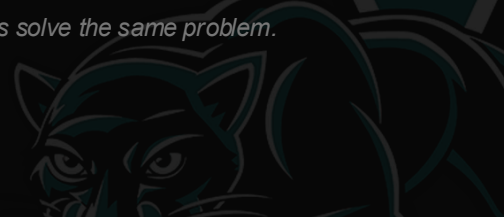
03

IT CHANGES WHAT LOOKS SHARP

Aperture controls depth of field too. That is all of Part 2.



Your eye and your lens solve the same problem.





01

LOOK FOR THE BLADES

Inside every lens is a ring of thin metal blades called the diaphragm.

02

THEY SLIDE, THEY DO NOT SWAP

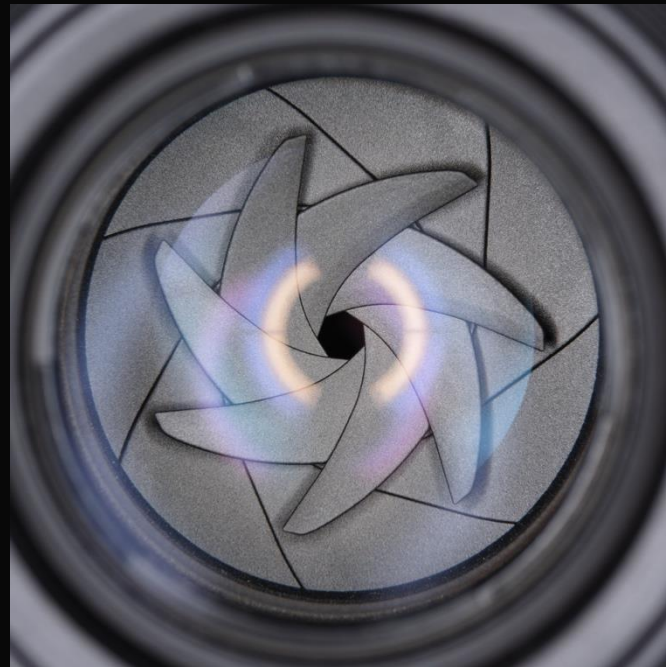
The blades slide across each other to open a bigger or smaller hole.

03

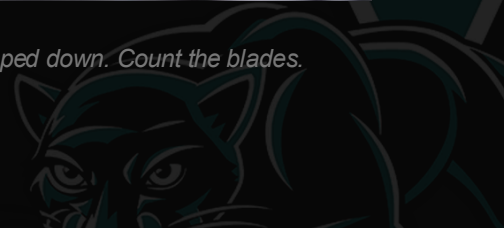
THE SHAPE MATTERS LATER

How many blades there are changes the look of out of focus lights. That is bokeh.

Take a lens off the camera and look through the back. You can watch the blades move.



The diaphragm, stopped down. Count the blades.





01

THE SLASH IS DOING REAL WORK

$f/4$ is written like a fraction on purpose. The opening is about one fourth as wide as the lens is long.

02

SO $f/2$ IS A BIGGER SLICE

One half is bigger than one fourth. That is why $f/2$ is wider than $f/4$.

03

THAT IS THE WHOLE TRICK

Once you see it as a fraction, the numbers stop feeling backwards.



Say it out loud as a fraction. "f four" becomes "one fourth" and the size makes sense. Every lens has its focal length printed on it, and the f-number is measured against that length, which is why $f/2$ on a short lens and $f/2$ on a long lens let in the same amount of light.



THIS IS THE ONE THING TO MEMORIZE TODAY

Small number, big opening

f/1.8 is a wide opening. Lots of light. Good for dark rooms and blurred backgrounds.

Big number, small opening

f/16 is a tiny opening. Very little light. Good for bright sun and keeping everything sharp.

It feels backwards because the number is the bottom of a fraction. A bigger bottom number always means a smaller slice. One sixteenth is smaller than one half.

REMEMBER THIS ONE LINE

Small f-number means big opening. If you take nothing else from today, take that.



SEE THE OPENING CHANGE

CONTENT / 08 / 20



f/1.8



f/2.8



f/4.0



f/5.6



f/16



f/22

Same lens, five f-stops. Nothing changes except how much light can get through.

LEFT TO RIGHT: WIDE OPEN TO STOPPED DOWN



WIDE OPEN

STOPPED DOWN



MORE LIGHT

LESS LIGHT

These are the full stops. Every step to the right cuts the light in half. Every step to the left doubles it.

Your camera moves in one third stops, so you will also see f/3.2 and f/4.5 in between. Those are fine. The full stops above are the ones to memorize.

COUNT IT YOURSELF

f/4 to f/8 is two full stops, so f/8 gets one quarter of the light. f/4 to f/2 is one stop wider, so f/2 gets twice as much.



01

ONE STOP WIDER DOUBLES THE LIGHT

Going from f/5.6 to f/4 lets in twice as much light.

02

ONE STOP SMALLER HALVES IT

Going from f/4 to f/5.6 cuts the light in half.

03

STOPS ARE THE SHARED LANGUAGE

Aperture, shutter speed and ISO are all measured in stops.

HALF IS A BIG CHANGE

Two stops is a quarter of the light. Three stops is an eighth.

f/2.8 to f/2

One stop wider. Twice the light hits the sensor.

f/2.8 to f/5.6

Two stops smaller. Only one quarter of the light.

Count the steps on the scale to count the stops.

You are at f/5.6 and the photo is too dark. Open to f/4 and you have doubled the light. Open again to f/2.8 and you have four times what you started with.



Every photo needs a certain amount of light to look right. Aperture is one of three ways to control it. Change one and you have to change another to stay balanced.

Aperture

How wide the lens opens. Also controls depth of field.

Shutter speed

How long the sensor is exposed. Also controls motion blur.

ISO

How sensitive the sensor is. Also controls picture noise.

Photographers call these three the exposure triangle. Open the aperture one stop and you can use a shutter speed one stop faster.





SET THE CAMERA TO FULL MANUAL

STEP 1 / 12 / 20

01

TURN THE MODE DIAL TO M

M is full manual. You set the aperture, shutter, and ISO.

02

SET YOUR APERTURE FIRST

Turn the main dial to pick the f-stop the photo needs.

03

THEN SHUTTER SPEED AND ISO

Balance the meter at zero, and keep the shutter at 1/60 or faster.



Everything you need is on one screen.



WATCH YOUR SHUTTER SPEED

WARNING / 13 / 20

UNDER 1/60 YOUR PHOTO WILL BLUR IN YOUR HANDS

Your photo comes out blurry and it is not the focus, it is camera movement.

The safe number

Keep your shutter speed at 1/60 or faster when the camera is in your hands.

If it drops below

Open the aperture wider, raise the ISO, or put the camera on a tripod.

Build the habit: glance at the shutter speed every time you change the f-stop. It takes half a second and it saves the photo.

In manual nothing moves unless you move it. Set the aperture for the look you want, then set the shutter speed to match the light.



THE THREE LENSES IN OUR KIT

REFERENCE / 14 / 20

RF-S 18-45MM

f/4.5 to f/6.3

The kit zoom. Wide to short telephoto. Your everyday lens.

RF 50MM

f/1.8

The prime. Does not zoom. By far the widest opening we own.

RF-S 55-210MM

f/5 to f/7.1

The long zoom. Reaches far. The smallest maximum opening of the three.

THE HIGHLIGHTED NUMBER IS THE MAXIMUM APERTURE

It is the widest that lens will ever open, and it is printed on the front of the barrel.



01

LOOK AT F/4.5 TO 6.3 AGAIN

That is not a range you choose. It is what the widest opening becomes as you zoom.

02

ZOOMED OUT YOU GET F/4.5

At 18mm the lens can open to f/4.5, its widest.

03

ZOOMED IN YOU ONLY GET F/6.3

At 45mm the widest it will go is f/6.3. You just lost light by zooming.

THIS ONE SURPRISES PEOPLE

Zoom in during a dark gym and the camera quietly takes light away from you.

18-45mm zoom

f/4.5 at the wide end. f/6.3 at the long end.

50mm prime

f/1.8 at every focal length, because it has only one.

Primes hold their maximum aperture. Zooms often do not.





In the industry a lens is called fast or slow. It has nothing to do with focusing speed. It describes the maximum aperture.

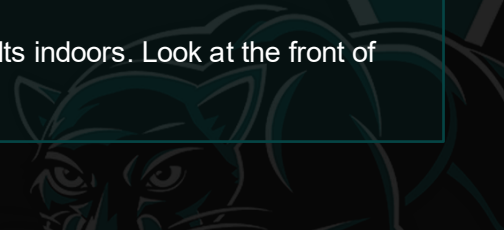
Fast lens

Wide maximum aperture, around $f/1.8$ or $f/2.8$. Lets in a lot of light, so the shutter speed can be fast.

Slow lens

Narrow maximum aperture, around $f/5.6$ or $f/6.3$. Needs more light, so the shutter speed has to be slower.

Our RF 50mm $f/1.8$ is a fast lens. Our 55-210mm at $f/7.1$ is a slow lens. Same camera, very different results indoors. Look at the front of any lens and you will find its maximum aperture printed there.





WHY THE 50MM IS WORTH GRABBING

CONTENT / 17 / 20

01

IT OPENS TO F/1.8

About three stops wider than the kit zoom at f/6.3. Eight times more light.

02

IT SAVES YOU INDOORS

Often the only lens in the kit that gets a clean photo without a tripod.

03

IT BLURS BACKGROUNDS BEST

A wide opening gives the shallowest depth of field in our kit.

04

YOU ZOOM WITH YOUR FEET

It does not zoom, so you walk closer or step back. That makes you frame on purpose.



RF 50mm f/1.8 STM. The fastest lens we own.





APERTURE IN LOW LIGHT

01

OPEN UP BEFORE YOU RAISE ISO

A wider aperture adds light for free. Raising ISO adds light and also adds noise.

02

THEN CHECK YOUR SHUTTER SPEED

If it is still under 1/60, you have run out of aperture and ISO is the next move.

03

THEN THINK ABOUT SUPPORT

A tripod or a steady surface lets you use a slow shutter speed without blur.

Order of operations indoors: open the aperture, raise the ISO, add support. In that order.





CHECK YOURSELF

STUDY / 19 / 20

01

WHICH LETS IN MORE LIGHT, F/2.8 OR F/8?

02

YOU MOVE FROM F/4 TO F/5.6. DID THE LIGHT DOUBLE OR GET CUT IN HALF?

03

WHAT IS THE MAXIMUM APERTURE OF OUR RF 50MM PRIME?

04

YOUR KIT ZOOM READS F/4.5 TO 6.3. WHAT HAPPENS WHEN YOU ZOOM IN?

05

IN MANUAL MODE, WHICH THREE SETTINGS DO YOU CONTROL?

Answer all five out loud with a partner before you pick up a camera.





WHAT YOU KNOW NOW

RECAP / 20 / 20

Aperture is an opening

A ring of blades inside the lens that you open and close.

The number is a fraction

Small f-number means a big opening. Big f-number means a small one.

Stops are steps of light

One stop either doubles the light or cuts it in half.

Manual puts you in charge

You set the aperture, the shutter speed, and the ISO.

You also know our kit: f/4.5 to 6.3 on the 18-45mm, a fast f/1.8 on the 50mm prime, f/5 to 7.1 on the 55-210mm.





COMING UP IN PART 2

NEXT MODULE

Part 1 was about light. Part 2 is about focus. You will learn what depth of field is, how aperture changes it, and how focal length and your distance change it just as much.

Depth of field

What it is and how to see it in a photo.

Focal length

Why 210mm blurs more than 18mm at the same f-stop.

Your distance

Why stepping closer blurs the background too.

Bring your camera and bring questions. Part 2 is where aperture stops being a number and starts changing how your photos actually look.

